

Web based
Application

Flipped Classroom Project

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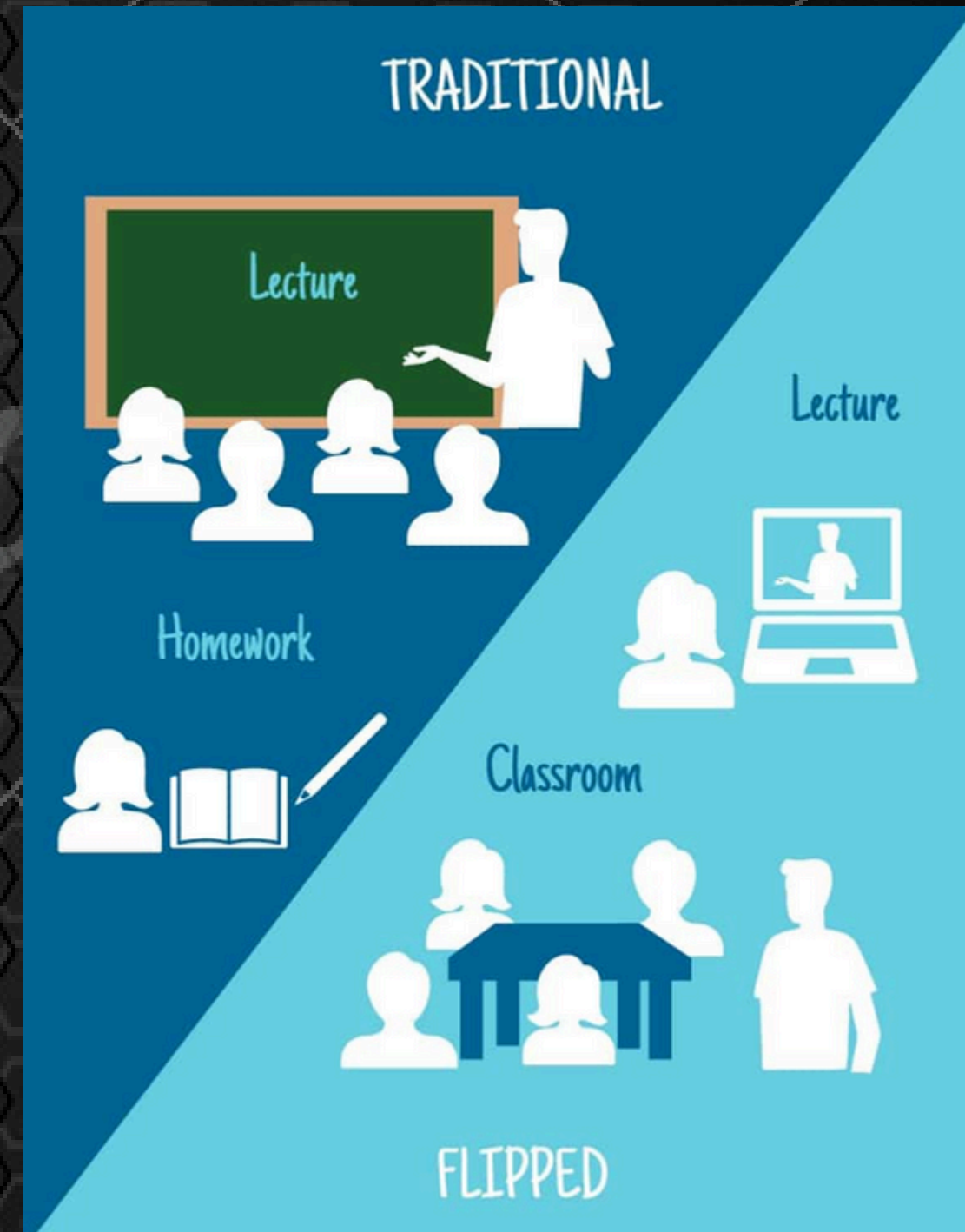
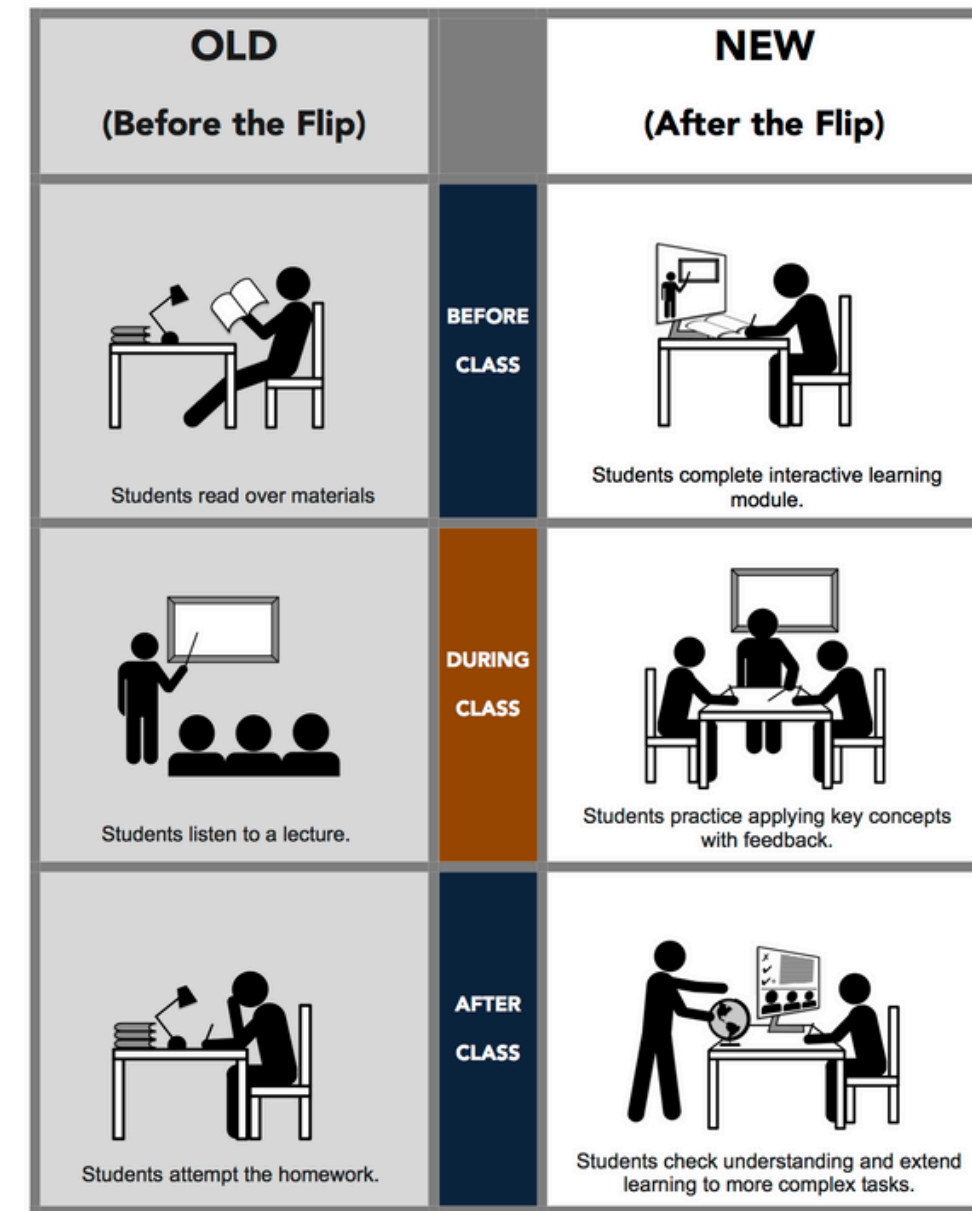


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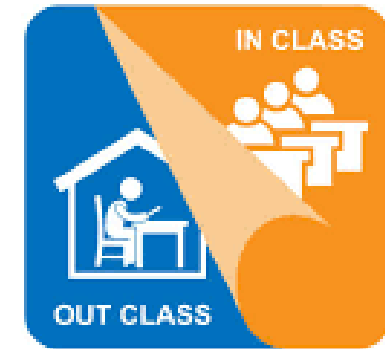
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Introduction

The Flipped Classroom is a modern teaching model where students learn instructional content at home through videos or materials and use classroom time for discussions, problem-solving, and interactive activities. This approach promotes active learning, deeper understanding, and more personalized teacher-student interaction by reversing the traditional lecture-homework structure.



Concept of Flipping



The flipped classroom model reverses traditional teaching by shifting direct instruction outside the classroom and using in-class time for active learning. The process typically begins with the teacher preparing digital content such as videos, reading materials, or slide decks. The core elements of the flipped model include:

- **Pre-Class Content Delivery:** Instructional materials made available online for students to study independently.
- **Active In-Class Engagement:** Classroom sessions are used for higher-order activities like analysis, synthesis, and application.
- **Student-Centered Learning:** Encourages autonomy and responsibility, allowing learners to control the pace and depth of their preparation.
- **Teacher as Facilitator:** The teacher guides, supports, and challenges students during class activities rather than delivering lectures.
- **Technology Integration:** A digital platform enables content sharing, assignment management, interaction, and performance tracking.

Process Flow Description



01

Admin configures the system by creating subjects and mapping users.

02

Teachers upload materials and assignments for their mapped subjects.

03

Students access materials, submit assignments, and participate in discussions via the chatroom.

04

- Teachers evaluate submissions and provide feedback.
- The system ensures data integrity, role-based access, and activity tracking throughout the workflow.

Project Objectives



Objective 01

Enhance Student Engagement and Active Learning
To shift from passive lecture-based learning to an interactive, student-centered approach.

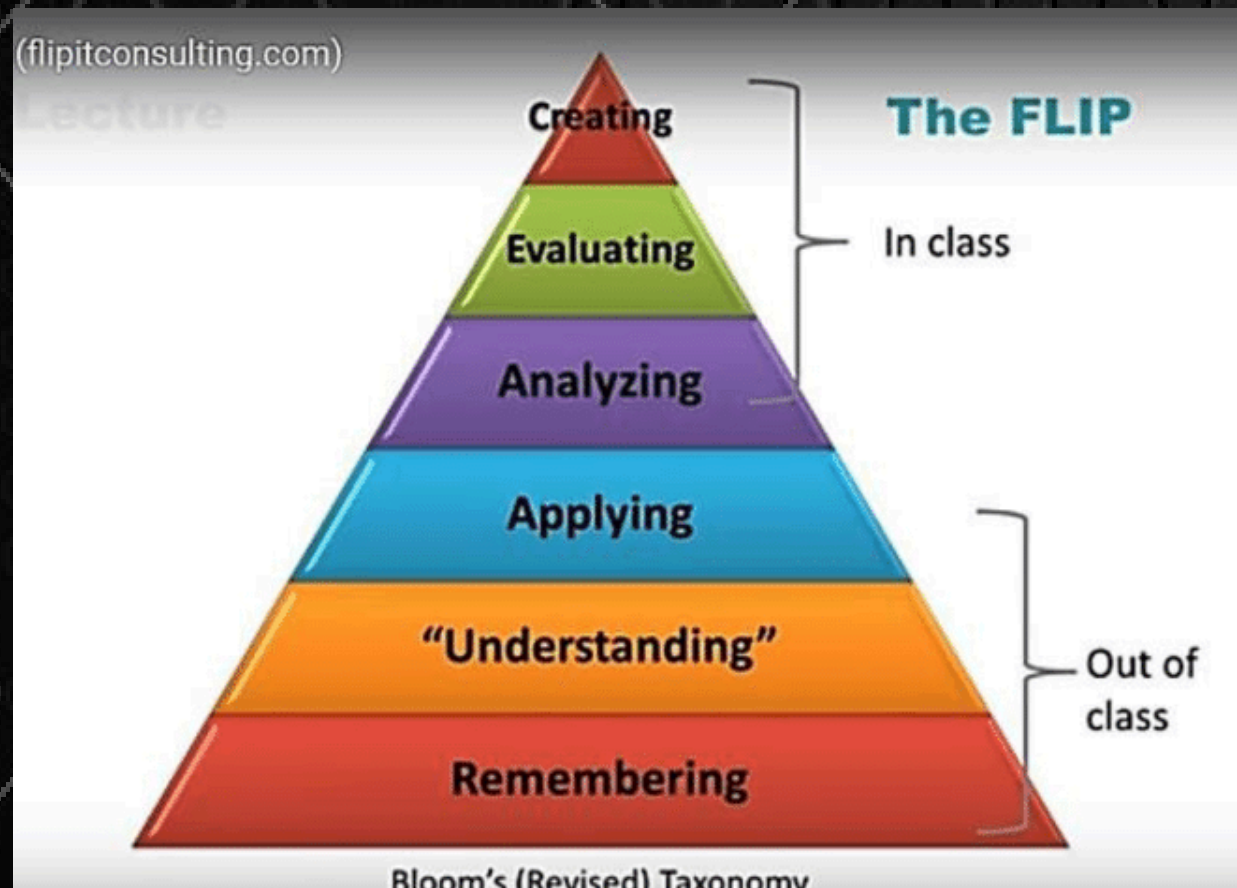
Objective 02

Provide Flexible and Accessible Learning Resources
To ensure students can access instructional materials anytime and anywhere, enabling self-paced preparation.

Objective 03

Streamline Teaching and Classroom Management
To equip teachers and administrators with a centralized digital platform.

Project Scope



Build a Multi-Role Academic System

Design a platform that distinctly supports students, teachers, and admins with tailored interfaces and permissions based on their roles.

Facilitate Flipped Learning Workflows

Implement key flipped classroom processes including pre-class content access, in-class interaction via chat, and assignment handling with grading support.

Ensure Scalable Class and Resource Management

Develop a backend system capable of managing dynamic class structures, user mappings, and digital resources for future scalability and maintenance.

Tech Stack



MVC Layers and Components



Model

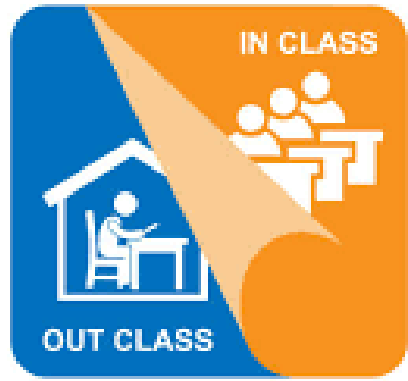
- Manages core data: users, classes, materials, assignments, grades, and chat logs
- Handles data validation, storage, and retrieval

View

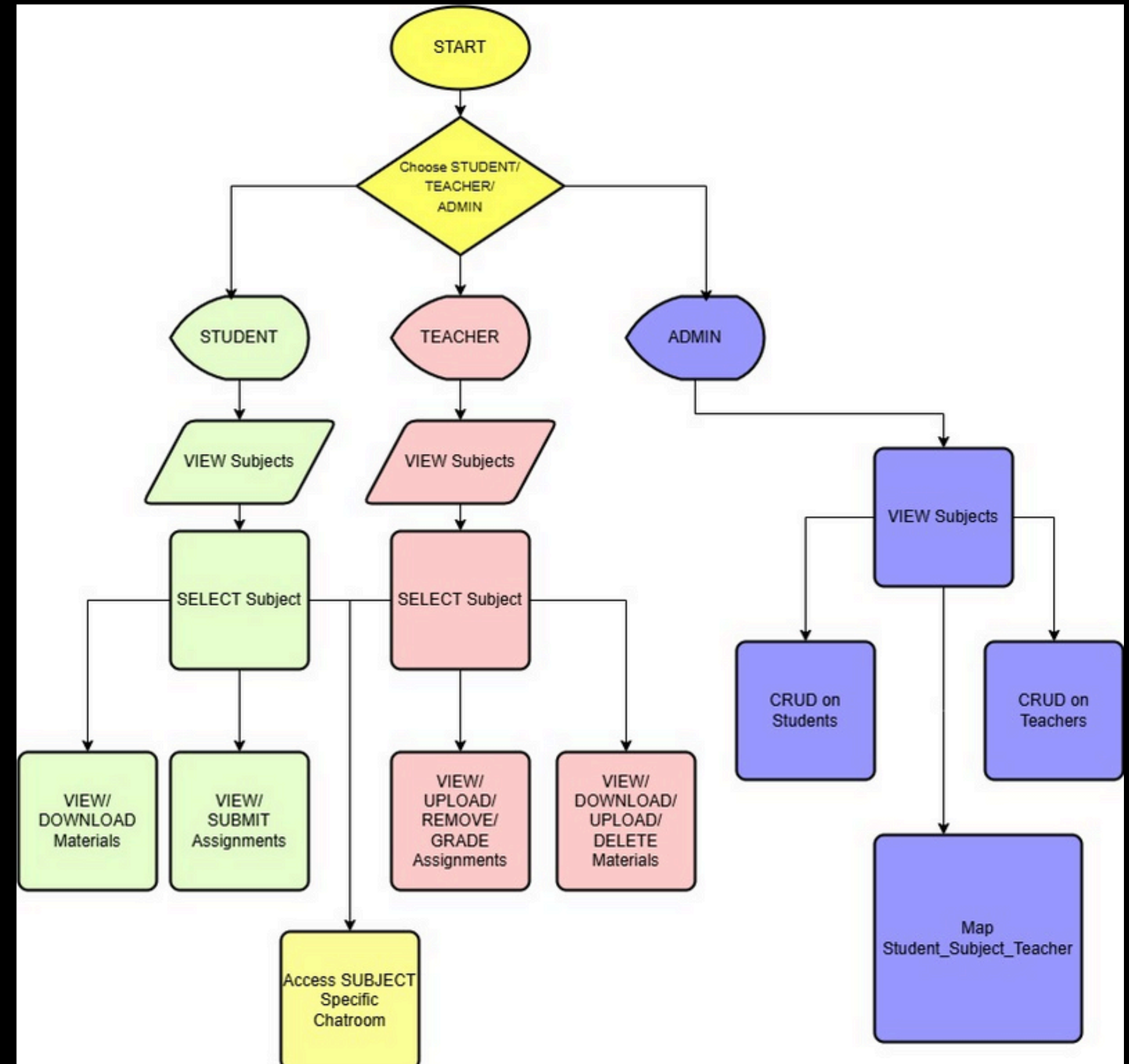
- Interfaces for Students, Teachers, and Admin
- Displays class materials, assignments, submissions, chatrooms, and dashboards
- Ensures a role-specific user experience

Controller

- Routes user requests (e.g., upload material, submit assignment, map class)
- Applies logic to decide what data to fetch or update
- Coordinates interaction between View and Model



Flowchart



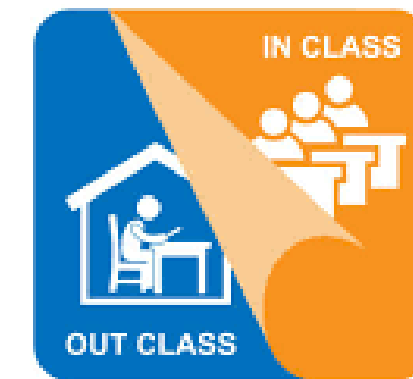
Results and Future Scope

01

Successfully developed a role-based flipped classroom system that enables material sharing, assignment handling, chat-based interaction, and class management for students, teachers, and admins.

02

Integrate features like video conferencing, performance analytics, and mobile app support to enhance accessibility, engagement, and real-time learning experiences.



Conclusion

The flipped classroom system redefines traditional learning by promoting active, student-centered engagement. Through streamlined content delivery, role-based access, and integrated tools, it creates a flexible and efficient digital learning environment.

Thank You!

